

KFC Sales Analytics

Data Cleaning, Analysis & Dashboard Report

Coverage: January 2023 – December 2025 · 8 stores · 3 countries

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Tools: Python (Pandas) · Power BI

1. Executive Summary

This report documents an end-to-end sales analytics project for a KFC-style restaurant chain, covering eight stores across Pakistan, the UAE, and the UK over a three-year period (2023–2025). The raw transactional dataset of approximately 104,500 records was cleaned and validated in Python, analysed for commercial insight, and delivered as an interactive one-page Power BI dashboard.

After cleaning, the dataset comprised 98,886 valid orders generating PKR 194.5 million in revenue. The single most important commercial finding is that bucket products dominate revenue: the Buckets category alone contributes roughly 36% of total revenue, led by the Chicken Bucket (8 pcs) at PKR 36.6M and the Boneless Bucket at PKR 32.9M. Revenue is otherwise remarkably stable across stores, cities, and time, and the average customer rating of 3.0/5 indicates clear room for improvement in customer satisfaction.

2. Data Overview

The raw dataset contained approximately 104,500 transaction rows with 14 fields: order identifiers, timestamps, store and location details, product and category, quantity, unit price, total amount, payment method, order type, customer age, and customer rating.

As is typical of operational data, the raw file contained substantial quality issues, including:

- **Missing values:** 4,168 missing cities, 8,288 missing ratings, 5,240 missing customer ages, and further gaps in payment method, order type, category, and order date.
- **Duplicates:** over 2,000 fully duplicated rows and roughly 4,500 duplicated order IDs carrying conflicting values.
- **Inconsistent formatting:** mixed casing and typos in city and product names (e.g. “Lahor”, “ZINGER BURGER”), inconsistent payment and order-type labels, and trailing whitespace in store IDs.
- **Invalid values:** impossible customer ages (e.g. 999), ratings outside the 1–5 scale, extreme quantities, negative sales amounts, and totals that did not equal quantity × unit price.
- **Mixed date formats:** around 10% of order dates stored in inconsistent formats.

3. Data Cleaning Methodology

Cleaning was performed in Python (Pandas) as a documented, repeatable pipeline. The guiding principle: recover what can be recovered from logical relationships, fill honestly what cannot, and drop only records that are both unrecoverable and essential.

Approach by issue

- **Text standardisation first:** all city, product, payment and order-type variants were mapped to canonical values, enabling reliable lookups in later steps.

- **Relationship-based recovery:** missing cities were recovered from store ID (each store belongs to exactly one city) and missing categories from product name — recovering 100% of these values without guesswork.
- **Invalid value handling:** impossible ages and out-of-range ratings were set to null before any statistical imputation; rows with impossible quantities or negative totals were removed.
- **Honest imputation:** missing payment methods and order types were labelled “Unknown” rather than filled with the most common value; missing ages were imputed with the median; missing ratings were deliberately left null, since absent ratings reflect customers who chose not to rate and imputing them would fabricate satisfaction data.
- **Integrity fixes:** full duplicates and duplicate order IDs were removed, total amounts were recomputed as quantity × unit price, and all dates were parsed to a single format.

The pipeline reduced the dataset from 104,500 to 98,886 rows (a 5.4% reduction), with every removal and imputation decision logged. The cleaned output feeds the Power BI dashboard directly.

4. Key Findings

4.1 Headline performance

Metric	Value
Total Revenue	PKR 194.5 million
Total Orders	98,886
Total Items Sold	297,126
Average Order Value	PKR 1,967
Average Customer Rating	3.0 / 5 (92% of orders rated)

4.2 Product and category insights

- **Buckets drive the business.** The Buckets category generates PKR 69.4M — 35.7% of all revenue — from only two products. Chicken Bucket (8 pcs) is the top earner at PKR 36.6M, followed by Boneless Bucket at PKR 32.9M.
- **A long tail of lower-value items.** Burgers (PKR 37.8M) and Meals (PKR 34.6M) form the second tier, while beverages and sides individually contribute the least revenue, with Pepsi (Regular) lowest at roughly PKR 3.0M. Low-price items remain strategically important as attachments to larger orders.

4.3 Geographic performance

- **Pakistan is the revenue base.** Pakistan contributes approximately 63% of revenue, the UK 25%, and the UAE 12%. This split is driven primarily by store count (five, two, and one store

respectively) rather than by differences in per-store performance — individual stores perform at a strikingly similar level (roughly PKR 24M each).

4.4 Customer behaviour and satisfaction

- **Balanced channels.** Dine-in, takeaway, and delivery orders occur in near-equal proportions, as do cash, card, and online payments. A small “Unknown” segment reflects records where these fields were not captured — itself a data-capture improvement point for operations.
- **Satisfaction is mediocre and measurable.** The average rating is 3.0/5 against a 92% rating response rate. On the dashboard this is visualised as a gauge against a target of 4.0, highlighting a clear satisfaction gap.

5. The Power BI Dashboard

All findings are delivered through a single-page interactive Power BI dashboard designed for at-a-glance executive reading, with the following components:

- **KPI cards:** total revenue, total orders, average order value, items sold, and average rating across the top of the page.
- **Monthly revenue trend:** a line chart on a zero-based axis showing revenue stability across the period.
- **Top 10 products:** a ranked bar chart highlighting the dominance of bucket products.
- **Category share:** a donut chart of revenue by product category.
- **Geography:** revenue by city, with a country-level map view.
- **Order mix:** order counts by channel (dine-in / takeaway / delivery).
- **Satisfaction gauge:** average rating (3.0) against a 4.0 target.
- **Interactive slicers:** year, city, and category filters that cross-filter every visual, allowing the client to self-serve answers (e.g. “Karachi only, 2024 only”) without new reports.

The model uses a dedicated date table for time intelligence, explicit DAX measures for every KPI, and a consistent visual identity (KFC red accent on a clean white canvas).

6. Recommendations

- **Protect and promote the bucket franchise.** With ~36% of revenue in two bucket SKUs, availability, quality and promotion of buckets should be treated as the top commercial priority; even small gains here move total revenue more than large gains elsewhere.
- **Grow average order value through attachments.** With a stable AOV of ~PKR 1,967, bundling low-cost sides and beverages with bucket and meal purchases is the most direct AOV lever.

- **Investigate the satisfaction gap.** A 3.0/5 average rating with high response volume is a reliable signal, not noise. A follow-up analysis linking ratings to store, product, and channel would identify where dissatisfaction concentrates.
- **Improve data capture at source.** The “Unknown” payment/order-type segment and the volume of raw-data defects justify tightening POS data-entry validation, which would improve every future analysis.

7. Limitations & Notes

This project was built on a synthetic practice dataset generated to mirror the structure and quality problems of real point-of-sale data. Absolute figures and the flatness of trends reflect that synthetic origin and should not be read as real KFC performance. The value of the deliverable lies in the pipeline itself: the cleaning methodology, the analytical approach, and the dashboard are production-ready and can be applied directly to real client data, where genuine seasonality, store differences, and customer patterns will surface through the same visuals.

All cleaning steps are documented and repeatable; rerunning the pipeline on updated data refreshes the dashboard without manual rework.